

Course Code	Name of the Course				
216U03C403	Communication Systems				
Teaching Scheme	TH	P	TUT	Total	
	03	0	0	03	
Credits Assigned	03	0	0	03	
Evaluation Scheme	Marks				
	LAB/ TUT CA	CA (TH)		ESE	Total
		IA	ISE		
		50	30		

Course pre-requisites:

Basic Electronics Circuits

Course Objectives:

The objective of the course is to introduce basic principles and techniques used in analog and pulse communication systems. The course addresses various concepts related to analog and pulse communication systems such as modulation, demodulation, transmitter, receiver and noise effect. The course also introduces analytical techniques to evaluate the performance of communication systems in time and frequency domains.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

- CO1.** Analyze and compare analog modulation schemes
- CO2.** Understand working of analog communication transmitter and receiver systems
- CO3.** Compare Pulse modulation systems and multiplexing techniques.
- CO4.** Understand noise effect on the performance of Communication Systems

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs.	CO
1	Fundamentals of Electronic Communication Systems		05	CO2 CO4
	1.1	Electromagnetic spectrum, Basic block diagram of communication system, Concept of modulation and demodulation		
	1.2	Sources of noise, signal to noise ratio, noise factor, noise figure, noise temperature and related numerical.		
2	Amplitude Modulation and Demodulation		11	CO1 CO2 CO4
	2.1	AM Signal: Waveform, mathematical representation, spectrum, modulation index, bandwidth, voltage distribution, power calculations and related numerical.		
	2.2	DSB-FC system: Modulation, low level and high level modulation; Demodulation: envelope detector, practical diode detector		
	2.3	DSB-SC system: Modulation, Balanced modulator, Ring modulator; Demodulation: Coherent Detector, Product detector		
	2.4	SSB system: Modulation: Filter method, Phase shift method, Third method; Pilot carrier SSB, Concept of VSB and ISB		
3	Basic Features of Angle Modulation and Demodulation		13	CO1 CO2 CO4
	3.1	Angle Modulation: Waveform, mathematical representation, modulation index, frequency and phase deviation, spectrum, bandwidth; Narrow Band FM, Wide Band FM		
	3.2	FM system: Modulation, Direct methods –Varactor diode modulator, FET reactance modulator; Indirect method – Armstrong's method; Noise triangle in FM; Pre-emphasis and de-emphasis; FM transmitter		
	3.3	PM system: Modulation, Direct and Indirect PM modulator; Relationship and comparison between FM and PM		
	3.4	FM Demodulation: Foster-Seeley discriminator, Ratio detector, FM demodulator using PLL; Comparison between AM, FM and PM		
4	Radio Receivers		06	CO2 CO4
	4.1	Characteristics of radio receiver, Choice of Intermediate Frequency, Image frequency, Concept of Automatic Gain Control and its type		
	4.2	Receivers: Super-heterodyne receiver DSB-FC, FM, Automatic Frequency Control		
5	Pulse Modulation and Demodulation		10	CO2

	5.1	Sampling theorem for low pass signals		CO3 CO4
	5.2	PAM, PWM and PPM systems: Modulation and demodulation, Applications		
	5.3	PCM: Transmitter and receiver; noise considerations in PCM		
	5.4	DM: Transmitter and receiver; noise considerations in DM; Adaptive delta modulation		
	5.5	Multiplexing: Time Division Multiplexing, Frequency Division Multiplexing		
Total			45	

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	B.P. Lathi, Zhi Ding	<i>Modern Digital and Analog Communication system</i>	Oxford University Press	Fourth Edition, 2017
2	Herbert Taub, Donald Schilling, Goutam Saha	<i>Principles of Communication Systems</i>	McGraw-Hill Electrical and Electronic Engineering Series	Fourth Edition, 2013
3	Wayne Tomasi	<i>Electronics Communication Systems</i>	Pearson education	Fifth Edition, 2008
4	R.P. Singh and S.D. Sapre	<i>Communication systems Analog and Digital</i>	McGraw Hill, India	Second Edition, 2017
5	Gorge Kennedy, Bernard Davis, SRM Prasanna	<i>Electronic communication system</i>	McGraw-Hill	Fifth Edition, 2012

*In addition to printed books, faculty can suggest (authentic) urls or e-books, e-contents etc.